

# 1.Introduction

The dataset chosen for this analysis consists of flight data from Goibibo, which is an online travel agency that acts as a mediator for booking travel tickets, focusing on routes between Bangalore and Hyderabad. This dataset includes various details about flights such as prices, airlines, departure and arrival locations, and flight classes. The primary objective of this analysis is to uncover insights regarding flight pricing, airline distribution, and overall trends in travel between these two cities. Specifically, aiming to determine median flight prices, analyze airline performance, and visualize the most popular destinations from Bangalore.

## 2. Data Description

The dataset contains several values out of which the following were selected

“from”: Originating city of the flight – (Categorical)

“to”: Destination city of the flight – (Categorical)

“airline”: The airline operating the flight – (Categorical)

“price”: The price of the flight to the passenger, excluding any discounts – (Numerical)

“class”: The passenger class of the specific route, economy, premium economy, business etc.  
– (Categorical)

Size of the dataset –The dataset consists of 300,261 rows and 13 columns, with 5 columns actively used in this analysis

## 3. Results

### Importing data -

Importing the data was done through the “import dataset” option in RStudio

The following code was also used

```
library(readxl)
goibibo_flights_data <-
read_excel("C:/Users/mohni/Downloads/goibibo_flights_data.xlsx")
```

### Missing Values -

In order to check for missing values and to remove them from the data while calculating results, “na.rm” argument is used, with the parameter of TRUE being assigned to them, in order to find values that are not applicable and remove them

## Data summarization -

The following data was summarized in the analysis

### - Measures of central tendency

In order to figure out the average price of flights from Bangalore to Hyderabad and vice versa, the “median” function was used as it is the most accurate method to find the central tendency in case of price, using the “mean” function would cause any outliers to heavily skew the results.

### - Aggregate data summarization

To find out which airline has the highest revenue per passenger/capita, aggregate function was used, this data was later visualized in the form of a scatter plot

## Exploring Data Distributions and Outliers

### - Calculating prices of flights separately based on passenger seating class

Airlines with business class seats were outliers in terms of median flight price due to the higher cost of business class tickets, therefore in order to account for the outlying data, economy class prices were calculated separately

- In order to visualise data related to median prices, airline distribution, revenue variations among airlines etc. many types of graphs were used such as barplots, line graphs, pie charts etc.

## Functions and Applying Control Structures

```
get_median_price <- function(airline_name
```

```
{
```

-“get\_median\_price” function was used in such a manner in order to find out the median price of flights per airline, instead of manually entering data separately for each airline, the function makes the process more efficient

- Control functions were put in place for varying airlines, passenger classes, routes, etc.

## 4. Interpretation and Conclusion

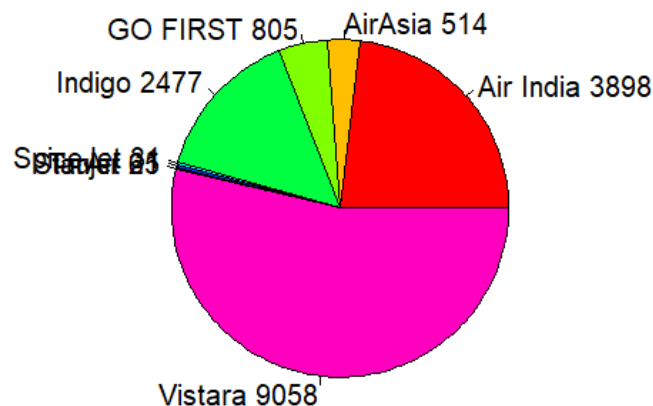
### i) Interpretation of median flight price

The median flight price from Bangalore to Hyderabad is 7891 rupees, whereas the median flight price from Hyderabad to Bangalore is 6924 rupees. The flights starting from Bangalore are more expensive due to the following reasons

- Supply and Demand:  
Bangalore, being a highly diverse metropolitan city has many people coming from neighbouring states for jobs, studies, etc. due to which there is higher demand for flights going out of Bangalore, making the airport more expensive for airlines, whose cost is passed on to the consumers
- ATF costs  
ATF, standing for Aviation Turbine Fuel is more expensive in Bangalore's main Kempegowda airport compared to Hyderabad airport, which reflects in ticket prices
- Taxes  
When booking an airline ticket out of Bangalore, a higher amount of tax is levied on the ticket when compared to Hyderabad, which increases prices. Furthermore with the continued development of Bangalore airport and the newly released T2, Kempegowda airport charges airlines operating out of it a higher amount.

## ii) Interpretation of Airline Distribution

### Airline Distribution: Bangalore <-> Hyderabad



- When analysing the distribution of flights between Bangalore and Hyderabad, it can be seen that the major of them are operated by Vistara and Air India, followed by Indigo. One limitation of this graph is that business class and economy class tickets both are counted, which can cause double counting in some cases as one plane can have both types of classes operating on the same route. In order to check airline distribution based on only economy class tickets, the required data can be filtered out by the following:

```
# Create a vector of target airlines
target_airlines <- c("Vistara", "Air India", "Indigo", "AirAsia", "GO FIRST",
"SpiceJet")

condition <- (goibibo_flights_data$from == "Bangalore" & goibibo_flights_data$to ==
"Hyderabad") |
              (goibibo_flights_data$from == "Hyderabad" & goibibo_flights_data$to ==
"Bangalore")

# Filter the dataset
filtered_flights <- goibibo_flights_data[goibibo_flights_data$airline %in%
target_airlines & condition, ]

# Create a contingency table for class counts
class_counts <- table(filtered_flights$airline, filtered_flights$class)

# Print the class counts
print(class_counts)
```

□

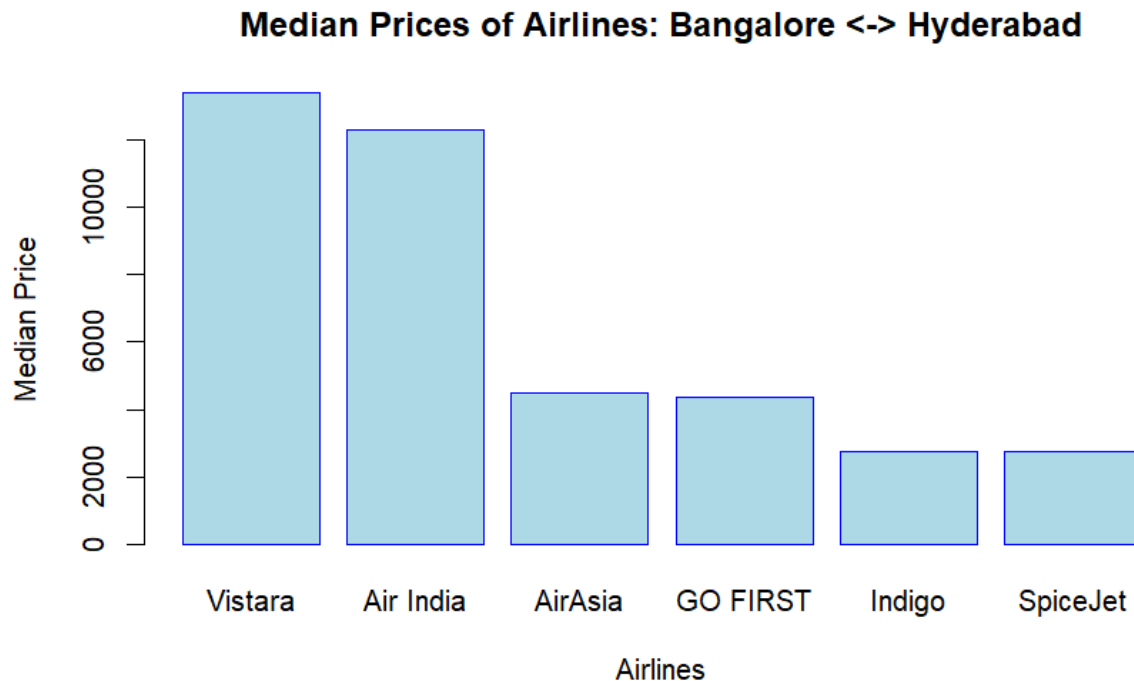
This gives the output of the following:

|           | business | economy |
|-----------|----------|---------|
| Air India | 1509     | 2389    |
| AirAsia   | 0        | 514     |
| GO FIRST  | 0        | 805     |
| Indigo    | 0        | 2477    |
| SpiceJet  | 0        | 31      |
| Vistara   | 4195     | 4863    |

With this it can be concluded that Vistara still leads the number of flights, however, not by an extremely large margin, and Indigo shifts to second place with Air India going to third.

### iii) Interpretation of median ticket price

An important statistic to consider when analysing flight routes is the median ticket price. In the beginning the median ticket price from Bangalore to Hyderabad and Hyderabad to Bangalore was calculated, this data can be calculated for varying airlines using a function along with a for loop. Plotting this data on a histogram, we get the following output



-We can see that Vistara and Air India have a high median price, compared to other airlines such as AirAsia, GO FIRST, Indigo and SpiceJet. This might be due to Vistara and Air India offering a more refined travel experience, while the other airlines tend to focus on affordability. However, this explanation does not explain the high discrepancy between the prices. With Vistara and Air India being almost four times more expensive than their budget oriented counterparts.

-Upon inspection of data, it is seen that business class tickets are almost 10 times the cost of economy class tickets. This is unusual, as with other routes, the difference between an economy and a business class ticket is approximately 2-3 times.

```
[1] "Vistara Business Median Price: 57553"
[1] "Vistara Economy Median Price: 6526"
[1] "Air India Business Median Price: 57439"
[1] "Air India Economy Median Price: 6377"
```

- In order to see why, we can go to the Vistara airlines website and check the prices for a

business class ticket from Bangalore to Hyderabad, when doing so, we can see that there is a layover in Mumbai which is far away from both the departure and arrival, causing the extremely high discrepancy between the economy and business class tickets compared to other routes

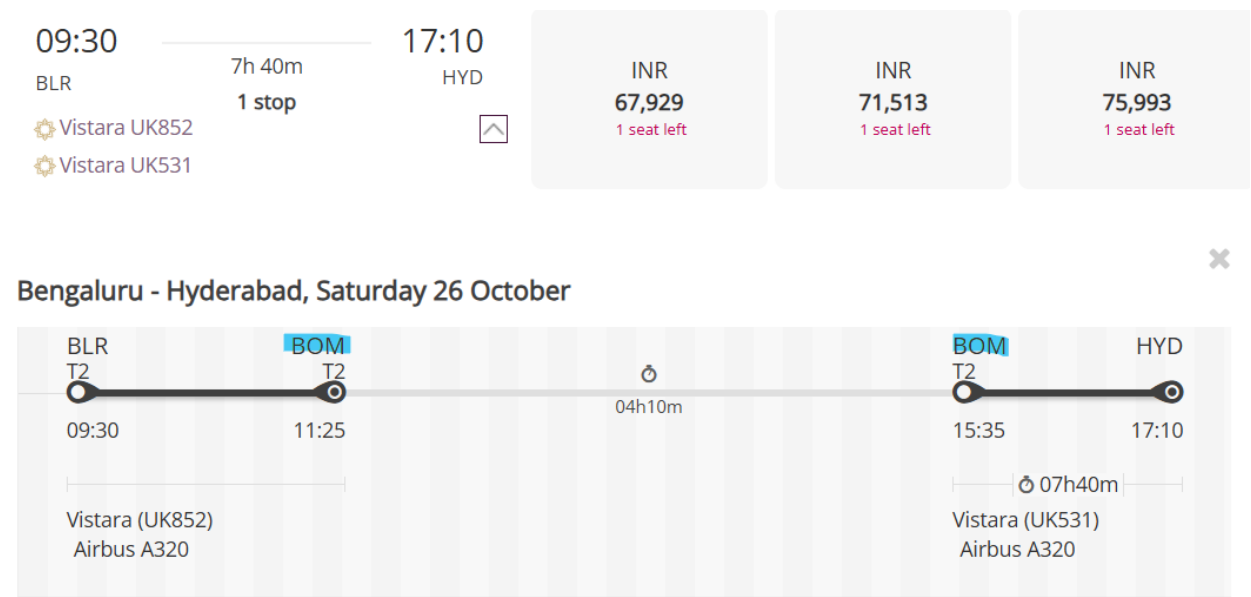


Fig - 1

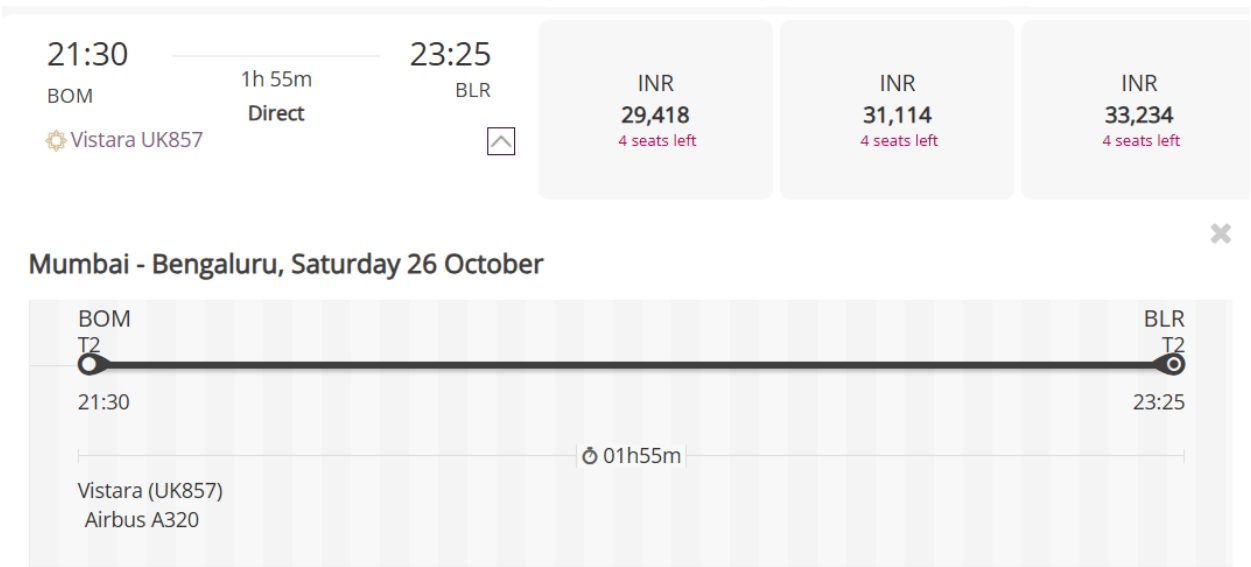


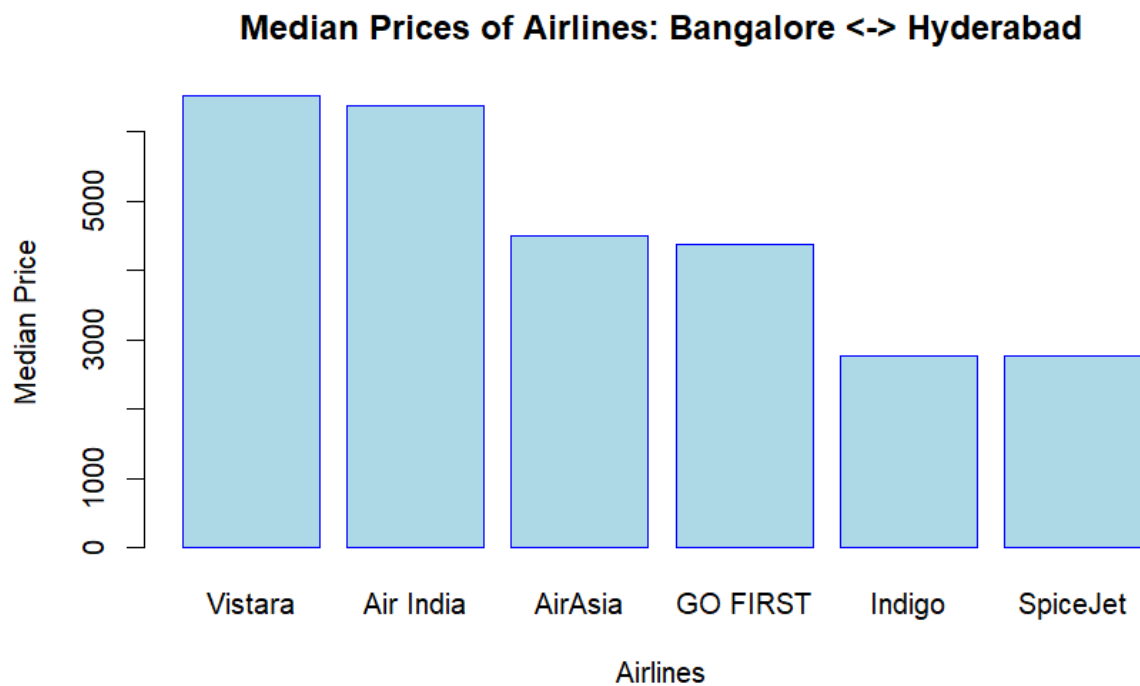
Fig - 2

In figure 1, we can see that the layover in Mumbai airport makes a flight ticket from Bangalore to Hyderabad more expensive than the same ticket from Bangalore to Mumbai, despite being a longer route

In order to get a more accurate comparison of median price, it is important to filter out airlines that offer business class tickets. When doing, the following output occurs

|           | business | economy |
|-----------|----------|---------|
| Air India | 1509     | 2389    |
| AirAsia   | 0        | 514     |
| GO FIRST  | 0        | 805     |
| Indigo    | 0        | 2477    |
| SpiceJet  | 0        | 31      |
| Vistara   | 4195     | 4863    |

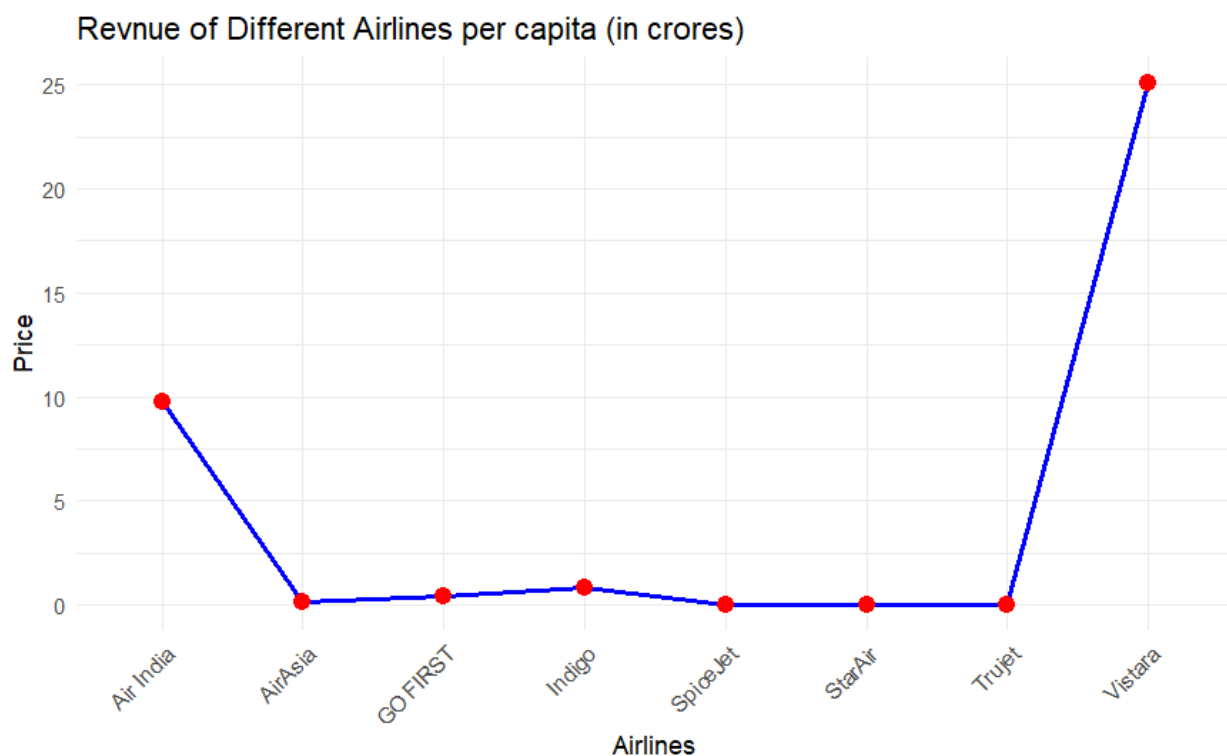
-It can be seen that Air India and Vistara are the only airlines offering business class between Bangalore and Hyderabad, by filtering out the business class prices and generated a histogram, we get the following output



Vistara and Air India still remain the most expensive airlines, however the price difference between them and the other airlines is a more accurate comparison, as business class tickets are filtered out

#### iv) Interpretation of Airline revenue

Using the ticket prices, we can estimate the revenue that airlines generate on this route, per capita. The actual revenue is much higher, however it cannot be calculated due to the limitations of the dataset, in that it does not account for the amount of tickets per flight. Using the available data, and calculating the revenue, plotting it on a line graph, the following output is generated:



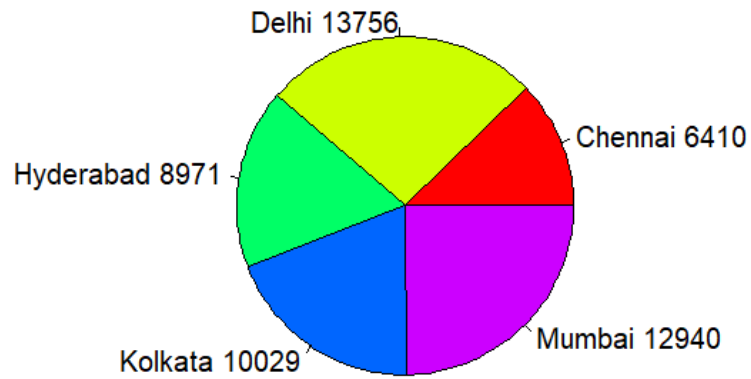
Once again, it can be seen that Air India and Vistara have the most estimated revenue, likely due to the high amount of flights along with large amount of business class passengers.

#### v) Interpreting Flight Destinations

Using the “to” column on the dataset, we can calculate the popularity of destinations of flights starting from Bangalore. Plotting this in a pie chart, we get the following:



## Flight Destinations from Bangalore



It can be seen that Chennai is the least popular destination, likely due to the less distance between both the cities, and the availability of other more efficient modes of transportation such as trains, buses and cars. Delhi and Mumbai are highest, likely due to various international flights using these airports as a connecting terminal.